

STA201 Assignment Solution 3-5 (Spring 2023)

1. The owner of a used car dealership is interested in researching how an automobile's age and selling price are related. Below is a sample of 12 used vehicles that the dealership sold during the course of the previous year. Here $X=10$.

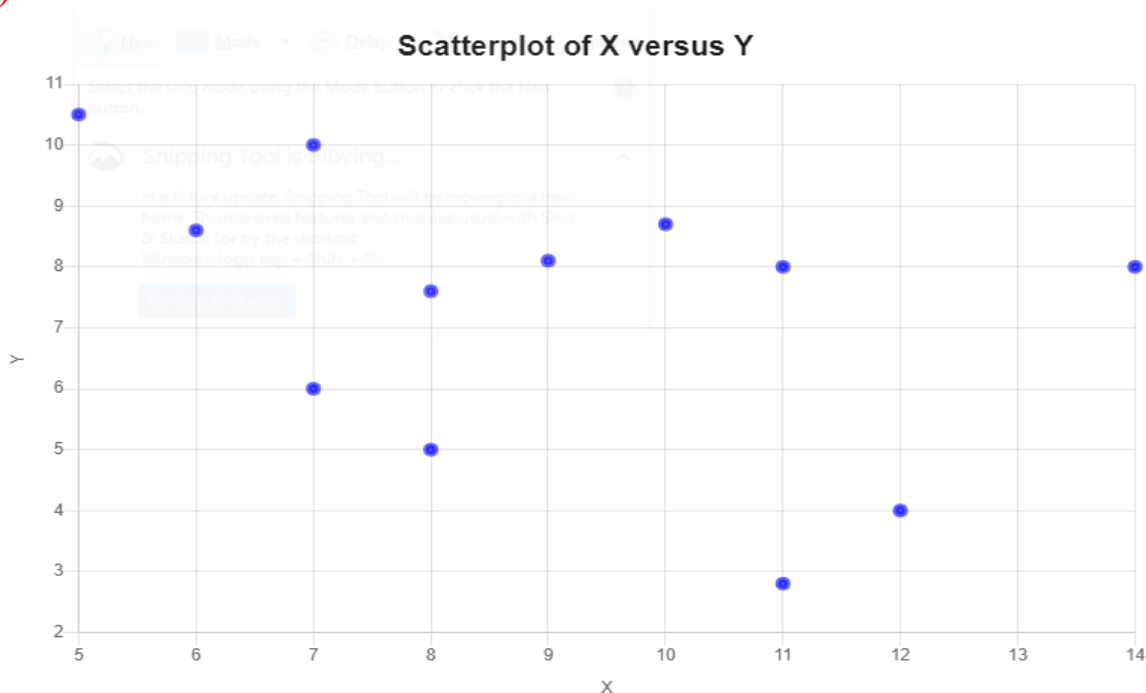
Age (years)	9	7	11	12	8	7	8	11	10	14	6	5
Price (thousand \$)	8.1	$X-4$	2.8	$X-6$	$\frac{X}{2}$	X	7.6	$\frac{X}{2}+3$	8.7	$X-2$	8.6	$X+\frac{1}{2}$

- Draw a scatter diagram and comment on the relation between the age of the car and its selling price.
- Determine the Pearson correlation coefficient and the coefficient of determination and interpret it.

Answer:

Age (years)	9	7	11	12	8	7	8	11	10	14	6	5
Price (thousand \$)	8.1	6	2.8	4	5	10	7.6	8	8.7	8	8.6	10.5

(a)



From the scatter diagram, a negative correlation between the age of a car and its selling price can be observed.

(b)

	Age (years) (X)	Price (thousand \$) (Y)	X^2	Y^2	XY
	9	8.1	81	65.61	72.9
	7	6	49	36	42
	11	2.8	121	7.84	30.8
	12	4	144	16	48
	8	5	64	25	40
	7	10	49	100	70
	8	7.6	64	57.76	60.8
	11	8	121	64	88
	10	8.7	100	75.69	87
	14	8	196	64	112
	6	8.6	36	73.96	51.6
	5	10.5	25	110.25	52.5
Sum:	108	87.3	1050	696.11	755.6

$$r = \frac{\sum_{i=1}^{12} (x_i y_i) - n \cdot \bar{x} \cdot \bar{y}}{\sqrt{\left(\sum_{i=1}^{12} x_i^2 - n \bar{x}^2 \right) \times \left(\sum_{i=1}^{12} y_i^2 - n \bar{y}^2 \right)}}$$

$$= -0.44$$

There is a moderate negative correlation between the age of a used car and its selling price.

$$r^2 = 0.1936 = 19.36\%$$

19.36% of the variation in the price of a used car can be explained by the age of the car.

2. The iodine value is the amount of Iodine necessary to saturate a sample of 100 g of oil. In the table below the first row states Iodine and second row states oil respectively.

132.0	129.0	120.0	113.2	105.0	92.0	84.0	83.2	88.4	59.0	80.0	81.5	71.0
46.0	48.0	51.0	52.1	54.0	52.0	59.0	58.7	61.6	64.0	61.4	54.6	58.8

- Determine the regression equation of 100 g of oil on the amount of Iodine. .
- Interpret the model.
- What is the predicted price when the amount of Iodine is 88
- Comment on the goodness of fit of the model.

Answer:

X	Y	XY	X²	Y²
132	46	6072	17424	2116
129	48	6192	16641	2304
120	51	6120	14400	2601
113.2	52.1	5897.72	12814.24	2714.41
105	54	5670	11025	2916
92	52	4784	8464	2704
84	59	4956	7056	3481
83.2	58.7	4883.84	6922.24	3445.69
88.4	61.6	5445.44	7814.56	3794.56
59	64	3776	3481	4096
80	61.4	4912	6400	3769.96
81.5	54.6	4449.9	6642.25	2981.16
71	58.8	4174.8	5041	3457.44
Sum= 1238.3	Sum= 721.2	Sum= 67333.7	Sum= 124125.29	Sum= 40381.22

- a. Determine the regression equation of 100 g of oil on the amount of Iodine.

Answer:

$$b_1 = \frac{n(\sum(x_i y_i) - (\sum x_i)(\sum y_i))}{n(\sum x_i^2) - (\sum x_i)^2}$$

$$= -0.22$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 76.43$$

$$\hat{y} = b_0 + b_1 x$$

$$\hat{y} = 76.43 - 0.22 x$$

- b. Interpret the model.

Answer:

$b_0 = 76.43$ means that the overall amount of oil will be 76.43, when the amount of iodine is zero.

$b_1 = -0.22$ means that the average overall amount of oil will decrease by 0.22 units, when the amount of iodine will increase by one unit.

- c. What is the predicted price when the amount of Iodine is 88.

Answer:

$$\begin{aligned}\hat{y} &= 76.43 - 0.22 x \\ &= 76.43 - (0.22 * 88) \\ &= 57.07\end{aligned}$$

- d. Comment on the goodness of fit of the model.

Answer:

$$\text{Total Sum of Squares (SST)} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - (y_i)^2/n$$

$$\text{Error Sum of Squares (SSE)} = \sum (y_i - \hat{y}_i)^2 = \sum y_i^2 - b_0 \sum y_i - b_1 \sum x_i y_i$$

$$\text{Goodness of fit of the model, } r^2 = 1 - \frac{SSE}{SST} \\ = 0.81$$

Comment: 81% variation in amount of oil saturated can be explained by variation in amount of iodine.

OR, Goodness of fit of the model, $R^2 = r^2 = (-0.90063)^2 = 0.81113$

3. Assume that the chances of a patient suffering from high blood pressure is 60%. It is also assumed that a course of meditation reduces the risk of high blood pressure by 45% and prescription of certain drugs reduces its chances by 55%. At a time, a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, the patient selected at random does not suffer from high blood pressure. Find the probability that the patient chose a course of meditation?

Let us define the events,

A_1 : Person is treated with meditation

A_2 : Person is treated with drugs

B : Person has high blood pressure

We have to find the probability that the patient followed a course of meditation given that the patient selected at random does not suffer from high blood pressure.

$$P(A_1|B') = \frac{P(B'|A_1) * P(A_1)}{P(B')}$$

It is given that, meditation and drug has equal probabilities,

$$P(A_1) = 0.5, P(A_2) = 0.5$$

We know the chances of having high blood pressure without any treatment is 60%

Meditation reduces the risk by 45% so the risk becomes 55% of the original.

Therefore, the probability of having high blood pressure if treated with meditation,
 $P(B|A_1) = 0.60 \times 0.55 = 0.33$

The drug reduces the risk by 55% so the risk becomes 45% of the original.

Therefore, the probability of having high blood pressure, if treated with drugs,
 $P(B|A_2) = 0.60 \times 0.45 = 0.27$

Therefore,

$$P(B) = P(B|A_1) * P(A_1) + P(B|A_2) * P(A_2) = (0.33 \times 0.5) + (0.27 \times 0.5) = 0.3$$

Finally,

$$P(A_1|B') = \frac{P(B'|A_1) \cdot P(A_1)}{P(B')} = \frac{(1-P(B|A_1)) \cdot P(A_1)}{1-P(B)} = \frac{(1-0.33) \cdot 0.5}{1-0.3} = \frac{0.085}{0.67} = 0.1269$$

4. A discrete random variable X has the following probability mass function

$$P(X = x) = \begin{cases} 2kx & x = 2, 4, 6 \\ k(x + 2) & x = 8 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant

- Show that $k = \frac{1}{34}$
- Find the exact value of $P(4 < x \leq 8)$
- Find the exact value of $P(2 < x < 4)$
- What is the expected value of the random variable X ?
- What is the variance of the random variable X ?
- Determine $Var(5 - 3X)$

ANSWER:

a) We know,

$$\sum f(x) = 1$$

Here,

$$\sum f(x) = (2k \times 2) + (2k \times 4) + (2k \times 6) + (k \times 8 + 2) = 34k$$

From the formula we can write,

$$34k = 1$$

$$\therefore k = \frac{1}{34} \text{ [SHOWED]}$$

b)

x	2	4	6	8
$P(X=x)$	$\frac{4}{34}$	$\frac{8}{34}$	$\frac{12}{34}$	$\frac{10}{34}$

$$\therefore P(4 < x \leq 8) = \frac{12}{34} + \frac{10}{34} = \frac{22}{34} = \frac{11}{17}$$

c) $P(2 < x < 4) = 0$

d) Expected value of X ,

$$E(X) = \sum x \times f(x) = 2 \times \frac{4}{34} + 4 \times \frac{8}{34} + 6 \times \frac{12}{34} + 8 \times \frac{10}{34} = 5.647$$

e) We know,

$$Var(X) = E(X^2) - [E(X)]^2$$

Here,

$$E(X^2) = \sum x^2 \times f(x) = 2^2 \times \frac{4}{34} + 4^2 \times \frac{8}{34} + 6^2 \times \frac{12}{34} + 8^2 \times \frac{10}{34} = 35.765$$

Therefore,

$$Var(X) = E(X^2) - [E(X)]^2 = 35.765 - 5.647^2 = 3.876$$

$$\begin{aligned}
 \text{f) } \text{Var}(5 - 3X) &= \text{Var}(-3x) \\
 &= (-3)^2 \times \text{Var}(x) \\
 &= 9 \times 3.876 = 34.884
 \end{aligned}$$

5. In each of 4 races, the Democrats have a 60% chance of winning. Assuming that the races are independent of each other, what is the probability that:

- (i) The Democrats will win 0 races, 1 race, 2 races, 3 races, or all 4 races?

Here, $n=4$, $p=0.60$, $q=0.40$

$${}^4C_0 * p^0 * q^{4-0} = 0.0256$$

$${}^4C_1 * p^1 * q^{4-1} = 0.1536$$

$${}^4C_2 * p^2 * q^{4-2} = 0.3456$$

$${}^4C_3 * p^3 * q^{4-3} = 0.3456$$

$${}^4C_4 * p^4 * q^{4-4} = 0.1296$$

- (ii) The Democrats will win at least 1 race.

$$P(\text{at least 1}) = P(X \geq 1) = 1 - P(\text{none}) = 1 - P(0) = 0.9744. \text{ Or,}$$

$$P(1) + P(2) + P(3) + P(4) = 0.9744.$$

- (iii) The Democrats will win a majority of the races.

$$P(\text{Democrats will win a majority}) = P(X \geq 3) = P(3) + P(4) = 0.3456 + 0.1296 = 0.4752.$$

6. Suppose on average, Nepal experiences 6 earthquakes per year.

- What is the mean number of earthquakes in Nepal in the first four month of a year?
- What is the probability that there'll be 7 earthquakes in Nepal in the next two years?
- What is the probability that there'll be at least 9 earthquakes in Nepal in 2024?

ANSWER:

- a) Average number of earthquakes per 12 months, $\lambda_{12} = 6$

$$\text{So, average number of earthquakes per 4 months, } \lambda_4 = \frac{6}{12} \times 4 = 2$$

- b) $\lambda_{12} = 6$

$$\therefore \lambda_{24} = 6 \times 2 = 12$$

Let, X = Number of earthquakes

$$\therefore P(X = 7) = e^{-\lambda_{24}} \cdot \frac{\lambda_{24}^x}{x!} = e^{-12} \cdot \frac{12^7}{7!}$$

- c) $\lambda_{12} = 6$

Let, X = Number of earthquakes

$$\therefore P(X \geq 9) = 1 - P(X < 9) = 1 - P(X \leq 8) = 1 - \left(\sum_{x=0}^8 e^{-6} \cdot \frac{6^x}{x!} \right) = 0.1528$$